Problem set 7
Due Monday October 20 at 11pm

Exercise 7.1 (8 points). Does the following series converge or diverge? Prove your answers.

$$\sum_{n=2}^{\infty} \frac{1}{n(\log_2(n))^2}$$

Exercise 7.2 (Rudin 3.9; 8 points). Find the radius of convergence of each of the following power series.

(a) $\sum n^3 z^n$ (b) $\sum \frac{2^n}{n!} z^n$ (c) $\sum \frac{2^n}{n^2} z^n$ (d) $\sum \frac{n^3}{3^n} z^n$

Exercise 7.3 (8 points). Let $a_{m,n} = \begin{cases} m & \text{if } m = n, \\ -m & \text{if } n = m+1, \\ 0 & \text{otherwise.} \end{cases}$

(a)

Does the series $\sum_{m=1}^{\infty} \left(\sum_{n=1}^{\infty} a_{m,n} \right)$ converge? If so, to what value?

(b)

Does the series $\sum_{n=1}^{\infty} \left(\sum_{m=1}^{\infty} a_{m,n} \right)$ converge? If so, to what value?

Exercise 7.4 (10 points). Suppose that (x_n) and (y_n) are sequences in $\mathbb{R}$, with all $x_n > 0$ and $y_n > 0$, and for some $L \in (0, \infty)$ we have

$$\lim_{n \to \infty} \frac{x_n}{y_n} = L \,.$$

Prove that $\sum_{n=1}^{\infty} x_n$ converges if and only if $\sum_{n=1}^{\infty} y_n$ converges.

(This is sometimes called the "limit-comparison test" for convergence of series.)

Exercise 7.5 (not for credit; Rudin 3.24, in part). Let X be a metric space. X might or might not be complete. If X is not complete, it would be nice to know how to "fill in the holes" to make it complete. This exercise explains a way of doing so: it constructs a new complete metric space X^* which has X as a subset. This construction is called *completion*, and is one way to construct $\mathbb{R}$ from $\mathbb{Q}$.

We call two Cauchy sequences (p_n), (q_n) in X *equivalent* if $d(p_n, q_n) \to 0$. We write this relation as $(p_n) \sim (q_n)$.

(1) Prove that this is an equivalence relation, i.e.
   (a) Any Cauchy sequence (p_n) has $(p_n) \sim (p_n)$,
   (b) If $(p_n) \sim (q_n)$ then $(q_n) \sim (p_n)$,
   (c) If $(p_n) \sim (q_n)$ and $(q_n) \sim (r_n)$, then $(p_n) \sim (r_n)$.

(2) If $(p_n) \sim (p'_n)$ and $(q_n) \sim (q'_n)$, prove that
$$\lim_{n\to\infty} d(p_n, q_n) = \lim_{n\to\infty} d(p'_n, q'_n)\,.$$
   (Note that the limit does exist, by the result of an earlier exercise.)

Now we divide the set of Cauchy sequences in X into *equivalence classes*: any two elements of a given class P are equivalent, and elements of different classes P, Q are not equivalent. Let X^* be the set of all equivalence classes of Cauchy sequences in X.

Then, define a distance function Δ on X^* as follows: if (p_n) is in the class P, and (q_n) is in the class Q, then
$$\Delta(P, Q) = \lim_{n\to\infty} d(p_n, q_n)\,.$$
The previous parts show that this distance function is well defined.

(3) Prove that the distance function Δ makes X^* into a metric space.

(4) Prove that X^* with this distance function is complete.

(5) Consider the map $\phi : X \to X^*$ which maps any $x \in X$ to the class of the Cauchy sequence $(x, x, x, \dots)$. Prove that ϕ is injective and $\Delta(\phi(x), \phi(y)) = d(x, y)$.